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DATA205, Spring 2026

Final Project report

Project overview

As criminality and public safety remain among the major concerns in the community, understanding the root causes of criminal activities is a crucial exercise. Therefore, through exploratory data analysis, statistical examination, and visual interpretation, this project seeks to identify relationships between income level, population count, and crime rates through key questions such as Does income level guarantee a lower crime rate? Does higher population correlate with more criminal activity?

This project is important because addressing criminality effectively requires a clear understanding of the root causes, and data-driven insights allow stakeholders (law enforcement authorities, government, policy makers) to tailor their strategies.

The primary dataset used in this project originates from DataMontgomery. With over 315,000 observations spread across 30 variables, it displays information about incidents reported to Montgomery County Police Department. In addition, the project used socio-economic data collected from the U.S. Census Bureau that generates information about median household income and the population across every district in Montgomery County.

It is to mention that only population and income are considered in this analysis, meaning other factors driving criminality are excluded. These exclusions may limit the completeness of the analysis and are acknowledged as potential gaps.

To complete the analysis, several tools were used such as:

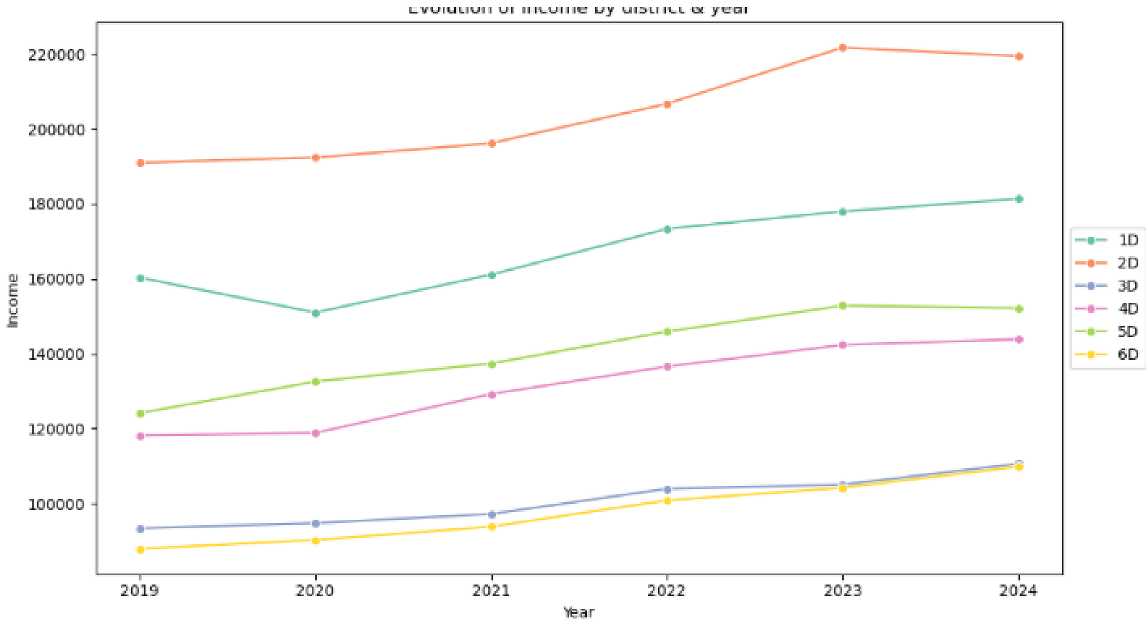
- Pandas: for data manipulation and cleaning
- Matplotlib & Seaborn: for data visualization
- Power BI: for data visualization
- Scipy: for correlation analysis

Prior to analysis, the raw dataset went through a cleaning process to ensure data quality consistency, and readiness. Records with missing values in key fields were identified and addressed. For instance, missing police district names were imputed using the mode. Furthermore, all duplicate records were removed to prevent misleading outcomes. Also, all date/time fields and other variables were standardized to an adequate data type for better processing. Most importantly, inconsistent district names (census district and police district) were reconciled to ensure accurate grouping.

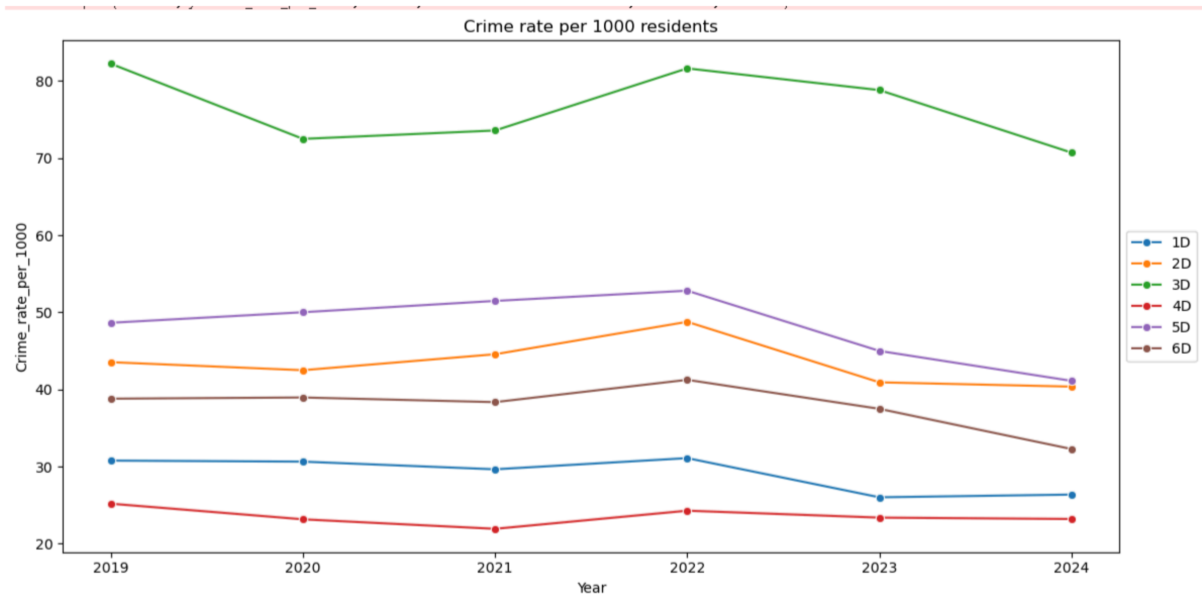
Summary/descriptive statistics

Within the dataset, crime is classified into three broad categories: Crime against property, crime against society, and crime against person. Crime against property, offenses that involve theft, damage or destruction of property without direct force or threat of force against a victim, constitute the largest proportion of incidents within the county with 170,000 incidents. It is followed by crime against society with 110,000, and crime against people with 40,000. Silver spring comes as the most criminal district within the county followed by Wheaton and Montgomery village to complete the top 3. Overall, all the incidents caused a total of approximately 323,000 victims across the county. Despite having a disparate count of incidents, the average daily crime across the districts is very approximate, ranging from 1.12 to 1.18 incidents per day. Silver spring has the highest record with an average of 1.18, followed by Wheaton and Rockville with 1.16.

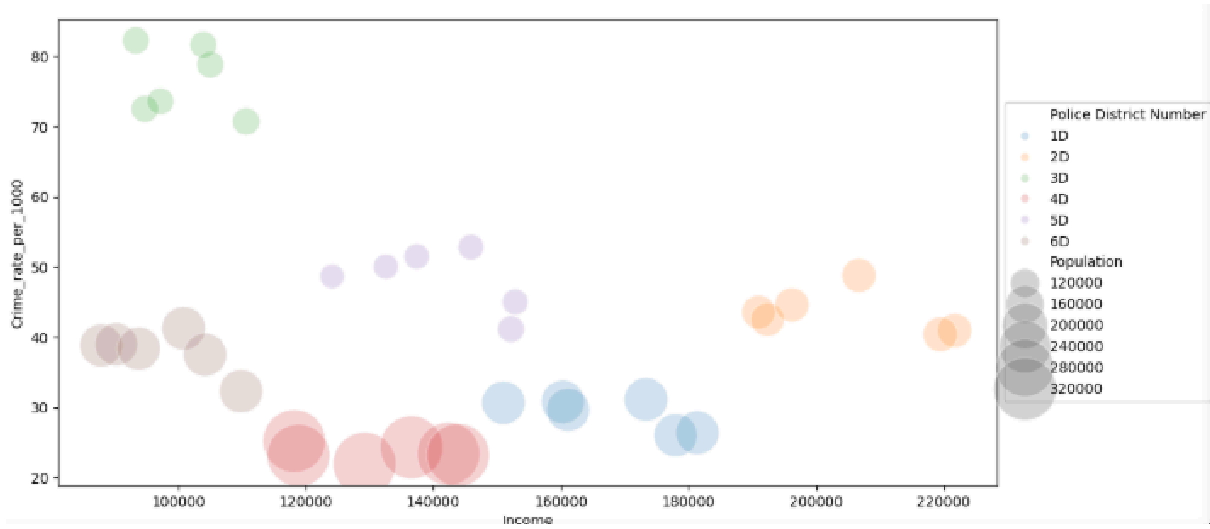
Crime/Income analysis



Ranging from 2019 to 2024, The 2nd district (Bethesda) has recorded the highest income level, rising from about 190,000 in 2019 to the upper 200,000 in 2023, to slightly decline in 2024 to 219,000. District 1(Rockville) follows as the second highest income level, growing from about 160,000 in 2019 to 182,000 in 2024. However, District 3 and District 6 recorded the lowest income levels moving closely around 90,000 in 2019 to about 110,000 in 2024. Overall, there's observable income growth, while the income gap remains large and stable over the period.



This line chart tracks the annual crime rate per 1,000 residents across the districts from 2019 to 2024. District 3(Silver Spring) consistently records the highest crime rate. District 5 exhibits the second highest rate, fluctuating between 49 and 53 incidents per 1,000 residents. However, district 4 and district 1 record the lowest crime rates over the period with 4D ranging from approximately 22 to 26 per 1,000 and 1D from 26 to 31 per 1,000. All districts experienced their peak before 2022 to then decline after 2023.



District 3 and district 6, positioned in the lower income range, show higher crime rates, with values ranging between 70 and 80 incidents per 1,000 residents. These districts also present smaller bubble sizes, suggesting lower population. Conversely, District 2 and district 1, placed toward a higher end of the income axis, exhibit lower crime rates.

Correlation analysis

To examine the strength of the association between income, population, and crime rate, a spearman correlation analysis was conducted. This method was used because it does not require conformity to a normal distribution and it does not assume linearity.

Ho : Income and population have no relationship with crime rate

Ha : Income and population have a relationship with crime rate

correlation matrix	annual income	population
crime_rate_per_100	-0.211326	-0.901673
annual income	1.000000	0.085199
Population	0.085199	1.00000

Considering the coefficient table, population and crime rate share a negative association with a correlation coefficient of -0.901673 . In contrast, the relationship between income level and crime rate is significantly weaker, showing a negative association with a coefficient of -0.211326.

P-values matrix	Income	Population
crime rate per 1000	2.159995e-01	6.251207e-14
Income	0.00000000	0.621267
Population	0.621267	0.00000000

The relationship between population and crime rate returned a p-value of 6.251207e-14, which is far below the standard 0.05 threshold, gives enough evidence that this association is not attributed to random variation. Furthermore, the relationship between income and crime rate returned a p-value of 0.216, which exceeds the 0.05 threshold. It indicates that income does not constitute a statistically reliable predictor

Conclusion

This analysis investigated the relationship between crime and income and population size across Montgomery county. The descriptive findings consistently identified District 3(Silver Spring) as the most crime intensive area, which is among the lowest income level areas. On the other hand, District 2, which commands the highest income levels in the county, maintained a lower crime rate. Population size emerged as the most reliable variable for driving crime, as suggested by the spearman correlation. Meanwhile, income demonstrated a weaker association. Collectively, the findings suggest that crime distribution in Montgomery County is more influenced by population size and, to a lesser extent, lower household income. These findings, collectively indicate that population size is a more reliable correlate of crime intensity than income at the district level.

Future analysis incorporating a wider range of socioeconomic indicators would further clarify the contributions of income and population to crime distribution across the county.